DOMAIN: ARTIFICIAL INTELLIGENCE – RULE-BASED EXPERT SYSTEM

AI-Based Bank Multi-Loan Eligibility and Pre-Screening System

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Abstract

- Banks receive many loan applications, and manual initial screening can be time-consuming.
- This project develops an AI-based, rule-driven pre-screening system.
- Users enter their loan and financial details.
- The expert system validates the information.
- Rule-based inference evaluates the configured eligibility conditions.
- The system produces an understandable pre-screening result.
- The system explains why the result was generated.
- Applications are stored in SQLite.

Software Requirements

Software

- Python
- Flask
- HTML5
- CSS3
- JavaScript
- Bootstrap 5
- SQLite
- VS Code
- Web Browser

AI Components

- Rule-Based Expert System
- IF–THEN inference
- Knowledge Base
- Validation Engine
- Explanation Generator

Hardware

- Basic computer / laptop
- 4 GB RAM or above
- Internet browser

Full View of the Project

Application Flow

Register / Login › Home / Dashboard › Select Loan › Application Form › AI Evaluation › Result › Application History

Major Pages

Registration / Login
Secure account access

Main Dashboard
Entry point to the system

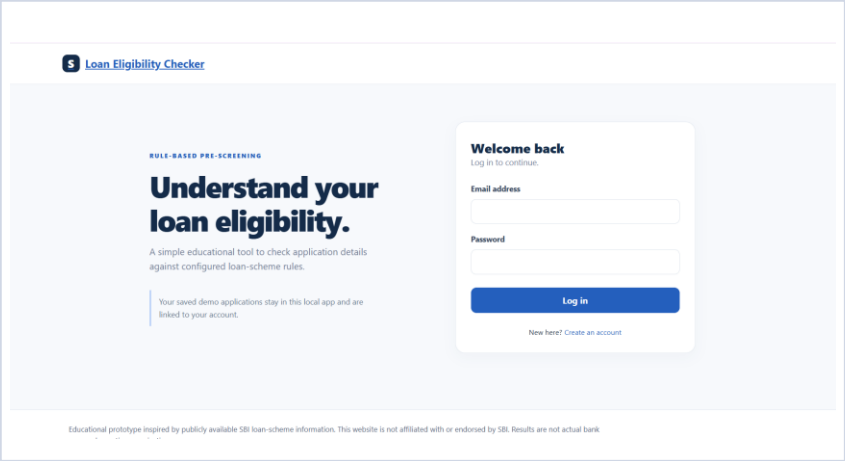
Loan Selection
Personal, Home, Car, Student

Application Form
Applicant and loan details

Result Page
Outcome with rule explanation

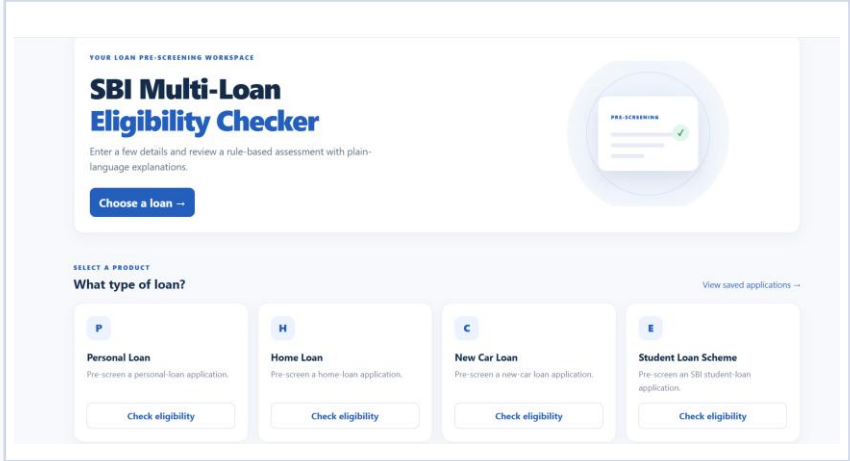
Application History
Previously saved applications

Full View – Login and Dashboard



Login Page

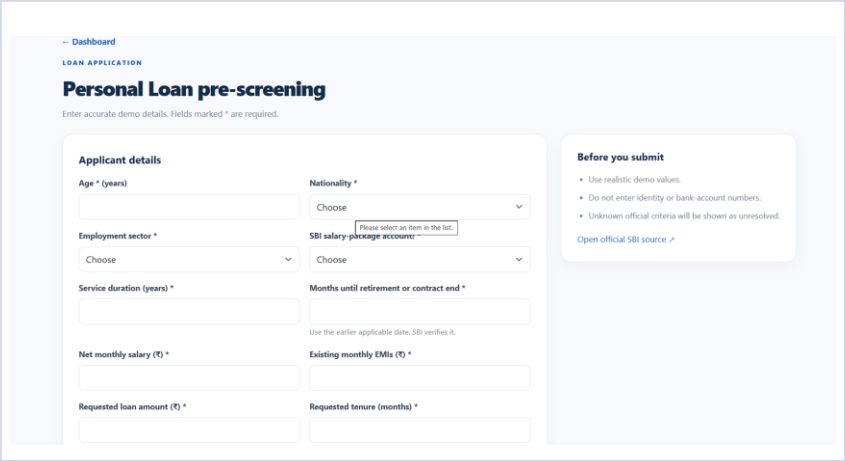
Users log in with email and password, or create a new account.

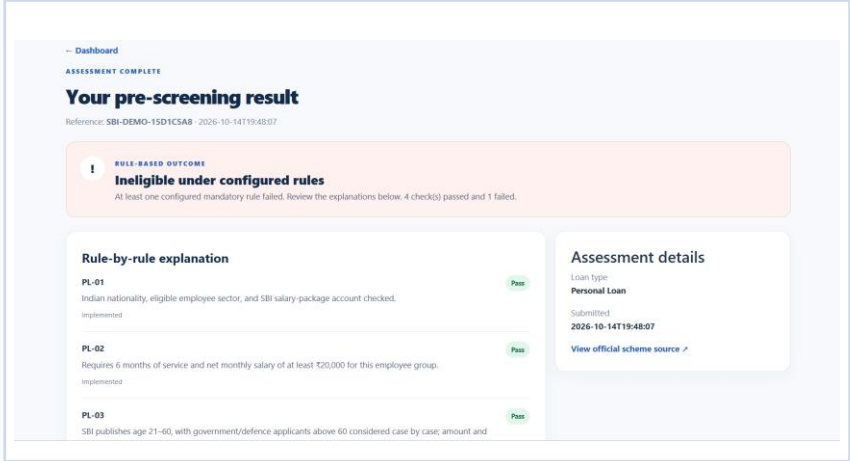


Main Dashboard – Loan Selection

Choose Personal, Home, New Car or Student loan to check eligibility.

Full View – Application and Result





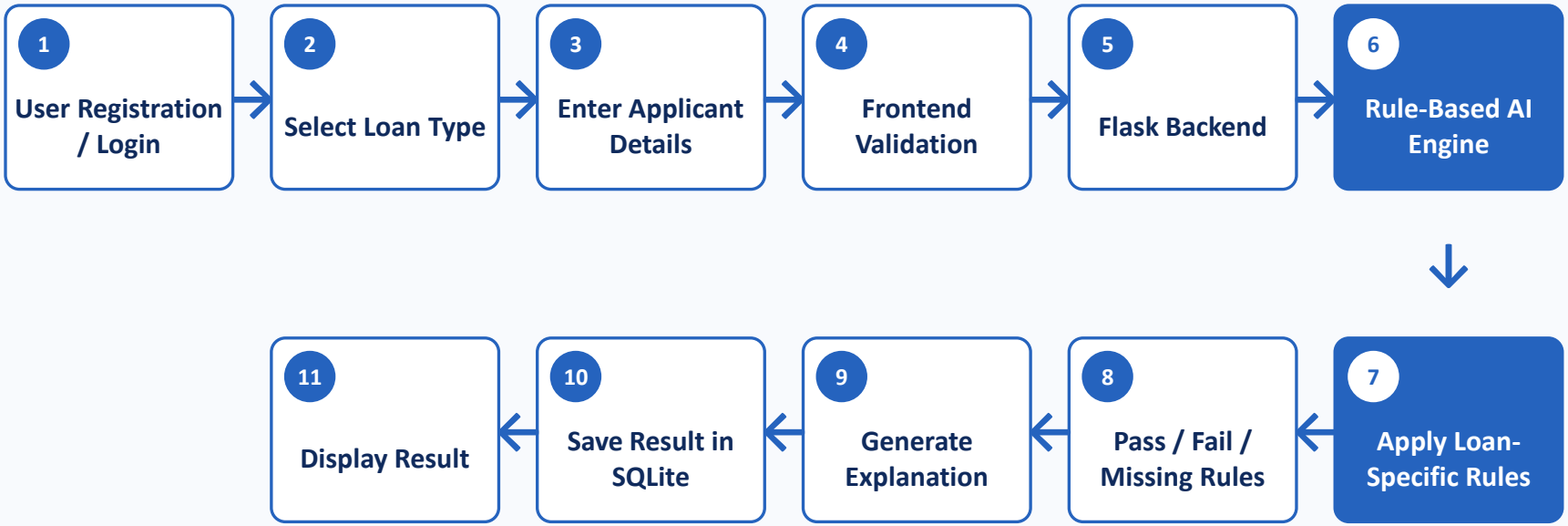
Application Form

Applicant and financial details entered with input validation.

Result Page

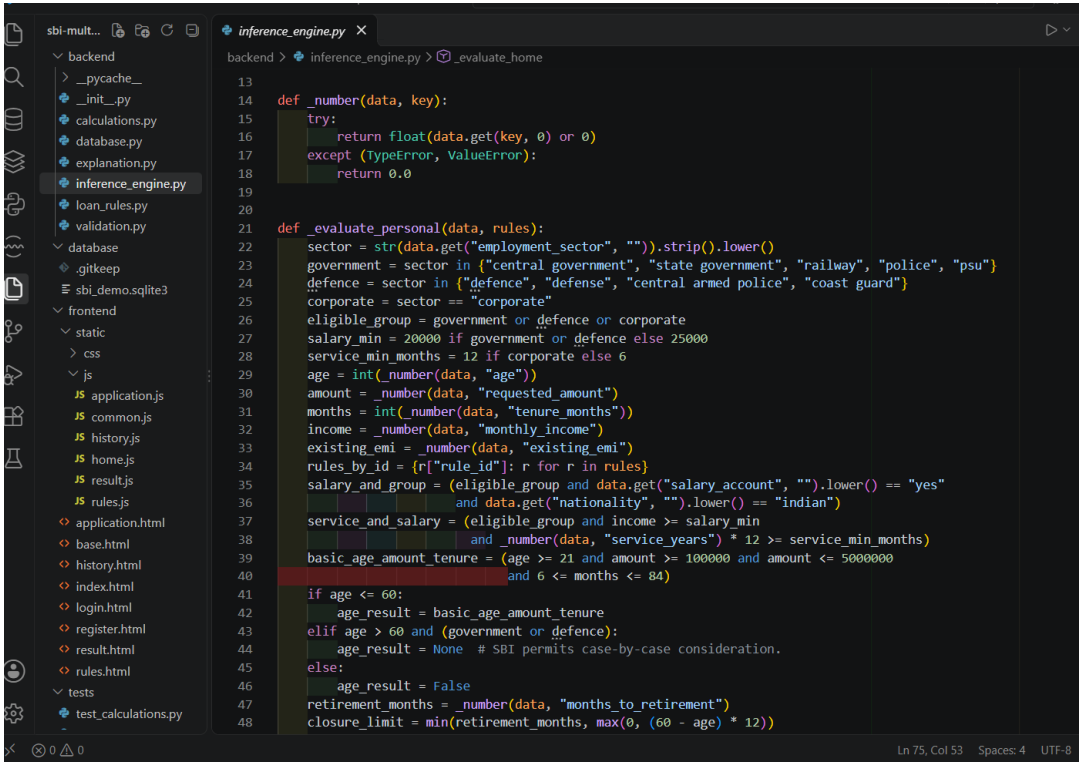
Pre-screening outcome with rule-by-rule Pass / Fail explanation.

Project Workflow



Steps 6–7 form the core of the expert system: IF–THEN rules are applied from the knowledge base.

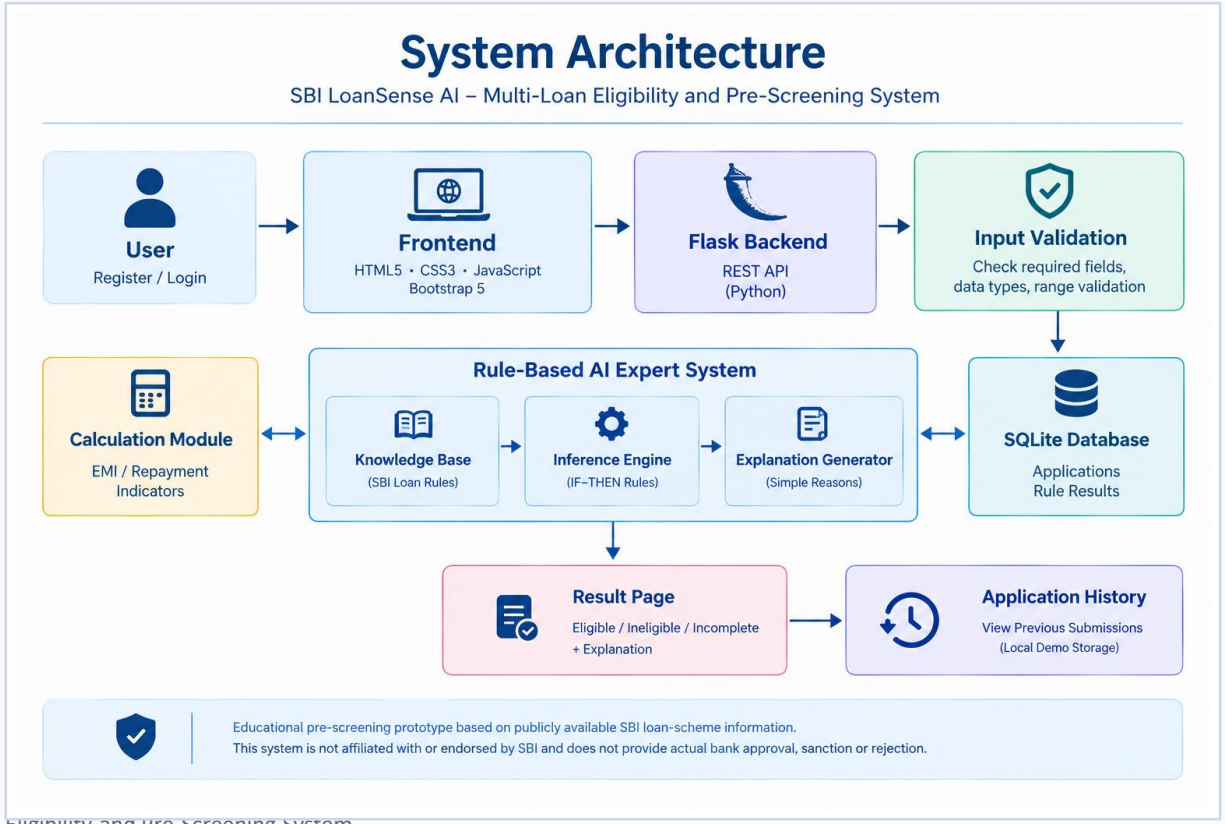
Sample Coding



inference_engine.py

- IF–THEN rules written in Python
- Checks sector, nationality and salary account
- Applies salary and service limits
- Validates age, amount and tenure
- Returns pass / fail for each rule

Architecture Diagram



Demo of Project

1. Register and log in
2. Choose a loan type on the dashboard
3. Fill in the application form
4. Submit for AI evaluation
5. View result with rule-by-rule explanation
6. Open application history

Live demonstration of the working website

Conclusion

The project demonstrates how a rule-based AI expert system can be used for automated loan pre-screening. By combining a knowledge base, validation, inference rules and explanation generation, the system provides consistent and transparent eligibility estimates. The project demonstrates a practical application of Artificial Intelligence in financial decision-support systems.

Thank you